

DRY-RUN GUIDE | 02_armstrong_number.cpp
Source: cpp\2nd_belt\02_armstrong_number.cpp

CORE BEHAVIOR

Counts decimal digits, raises each digit to that count using multiplication, and compares the sum to the original number.

REPRESENTATIVE INPUT / CALL

Input: '153'.

STEP TRACE

Digit count=3. Accumulate $3^3=27$, $5^3=125$, $1^3=1$; sum=153, equal to original.

EXPECTED OUTPUT / RETURN

'Armstrong'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | The input consists of a single integer N.
004 |
005 | Print "Armstrong" if the input number is an Armstrong number,
006 | otherwise print "Not Armstrong".
007 |
008 | Example:
009 | Input: 153
010 | Output: Armstrong
011 |
012 | 153 has 3 digits:
013 |  $1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$ .
014 | */
015 |
016 | #include <iostream>
017 | using namespace std;
018 |
019 | int main() {
020 |     int n;
021 |     cin >> n;
022 |
023 |     int original = n;
024 |     int digits = 0;
025 |     int temp = n;
026 |
027 |     if (temp == 0) {
028 |         digits = 1;
029 |     } else {
030 |         while (temp != 0) {
031 |             digits++;
032 |             temp /= 10;
033 |         }
034 |     }
035 |
036 |     long long sum = 0;
037 |     temp = n;
```

```
038 |  
039 |     while (temp != 0) {  
040 |         int digit = temp % 10;  
041 |         long long power = 1;  
042 |  
043 |         for (int i = 0; i < digits; i++) {  
044 |             power *= digit;  
045 |         }  
046 |  
047 |         sum += power;  
048 |         temp /= 10;  
049 |     }  
050 |  
051 |     if (sum == original) {  
052 |         cout << "Armstrong";  
053 |     } else {  
054 |         cout << "Not Armstrong";  
055 |     }  
056 |  
057 |     return 0;  
058 | }  
059 |
```